CEVE 421/521 Schedule

> **Note**
>
> Check the page for each reading to see discussion questions. You will often be encouraged to focus only on part of relevant papers.
>
> **Final Project:** Deliverables are due throughout the semester (see schedule below).
>
> **CEVE 521 students:** You will lead one Paper Discussion session. See Graduate Seminar Requirements.

Week	Date	What
1	Mon., Jan. 12	**Lecture:** Intro to Climate Risk Management [ ]
	Wed., Jan. 14	**Topical Discussion:** T. Frank [1] and S. Crawford [2]
	Fri., Jan. 16	**Lab 1:** Julia Setup & Hello World
2	Mon., Jan. 19	*Martin Luther King, Jr. Day (No Class)*
	Wed., Jan. 21	**Lecture:** Hazard, Exposure, Vulnerability [ ]
	Fri., Jan. 23	**Lab 2:** Intro to iCOW
3	Mon., Jan. 26	**Lecture:** Climate Variability, Extremes, and Change [ ]
	Wed., Jan. 28	**Paper Discussion:** S. Seneviratne *et al.* [3]
	Fri., Jan. 30	**Lab 3:** Sea-Level Rise Scenarios with BRICK / **Project:** Topic Proposal Due
4	Mon., Feb. 2	**Lecture:** Uncertainty & Monte Carlo [ ]
	Wed., Feb. 4	**Paper Discussion:** M. Zarekarizi, V. Srikrishnan, and K. Keller [4]
	Fri., Feb. 6	**Lab 4:** Monte Carlo Hazard Modeling
5	Mon., Feb. 9	**Lecture:** Risk Metrics (EAD, VaR, CVaR) [ ]
	Wed., Feb. 11	**Lab 5:** Probabilistic Risk Analysis
	Fri., Feb. 13	*No Class*

Week	Date	What
6	Mon., Feb. 16	**Exam 1** (Covers Module 1)
	Wed., Feb. 18	**Lecture:** Valuation & Benefit-Cost Analysis [ ]
	Fri., Feb. 20	**Paper Discussion:** K. Arrow *et al.* [5]
7	Mon., Feb. 23	**Lecture:** Optimal Static Decisions [ ]
	Wed., Feb. 25	**Paper Discussion:** D. van Dantzig [6]
	Fri., Feb. 27	**Lab 6:** Simulation-Optimization / **Project:** Memo 1 Due
8	Mon., Mar. 2	**Lecture:** Sensitivity & Value of Information [ ]
	Wed., Mar. 4	**Paper Discussion:** M. C. Runge, S. J. Converse, and J. E. Lyons [7]
	Fri., Mar. 6	**Lab 7:** Value of Partial Information
9	Mon., Mar. 9	**Lecture:** Trade-offs & Values [ ]
	Wed., Mar. 11	**Paper Discussion:** T. M. Daw *et al.* [8]
	Fri., Mar. 13	**Exam 2** (Covers Module 2)
	Mar. 16-20	*Spring Break*
10	Mon., Mar. 23	**Lecture:** Robustness & Deep Uncertainty [ ]
	Wed., Mar. 25	**Paper Discussion:** R. J. Lempert and M. E. Schlesinger [9] and S. H. Schneider [10]
	Fri., Mar. 27	**Lab 8:** Robustness Metrics / **Project:** Memo 2 Due
11	Mon., Mar. 30	**Lecture:** Exploratory Modeling & Scenario Discovery [ ]
	Wed., Apr. 1	**Paper Discussion:** S. Bankes [11]
	Fri., Apr. 3	**Lab 9:** Scenario Discovery
12	Mon., Apr. 6	**Lecture:** Sequential Decision Making [ ]
	Wed., Apr. 8	**Paper Discussion:** G. G. Garner and K. Keller [12]
	Fri., Apr. 10	**Lab 10:** Real Options & Dynamic Policy Search / **Project:** Memo 3 Due
13	Mon., Apr. 13	**Lecture:** Multi-Objective Optimization [ ]

Week	Date	What
	Wed., Apr. 15	**Paper Discussion:** J. R. Kasprzyk, S. Nataraj, P. M. Reed, and R. J. Lempert [13]
	Fri., Apr. 17	**Lab 11:** Pareto Fronts / **Project:** Slides Due
14	Mon., Apr. 20	**Final Project Presentations** (Teams 1 & 2)
	Wed., Apr. 22	**Final Project Presentations** (Teams 3 & 4)
	Fri., Apr. 24	**Final Project Presentations** (Teams 5 & 6)

Bibliography

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[2] S. Crawford, “Floods and Storms Are Ravaging the Jersey Shore. Why Do We Keep Building It Back?.” Accessed: Dec. 05, 2025. [Online]. Available: https://www.rollingstone.com/culture/culture-features/floods-storms-jersey-shore-beach-rebuilding-1235477927/

[3] S. Seneviratne *et al.*, “Weather and Climate Extreme Events in a Changing Climate,” *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, UK and New York, NY, USA, 2021. doi: 10.1017/9781009157896.013.

[4] M. Zarekarizi, V. Srikrishnan, and K. Keller, “Neglecting Uncertainties Biases House-Elevation Decisions to Manage Riverine Flood Risks,” *Nature Communications*, vol. 11, no. 1, p. 5361, Oct. 2020, doi: 10.1038/s41467-020-19188-9.

[5] K. Arrow *et al.*, “Determining Benefits and Costs for Future Generations,” *Science*, vol. 341, no. 6144, pp. 349–350, Jul. 2013, doi: 10.1126/science.1235665.

[6] D. van Dantzig, “Economic Decision Problems for Flood Prevention,” *Econometrica*, vol. 24, no. 3, pp. 276–287, 1956, doi: 10.2307/1911632.

[7] M. C. Runge, S. J. Converse, and J. E. Lyons, “Which Uncertainty? Using Expert Elicitation and Expected Value of Information to Design an Adaptive Program,” *Biological Conservation*, vol. 144, no. 4, pp. 1214–1223, Apr. 2011, doi: 10.1016/j.biocon.2010.12.020.

[8] T. M. Daw *et al.*, “Evaluating Taboo Trade-Offs in Ecosystems Services and Human Well-Being,” *Proceedings of the National Academy of Sciences*, vol. 112, no. 22, pp. 6949–6954, Jun. 2015, doi: 10.1073/pnas.1414900112.

[9] R. J. Lempert and M. E. Schlesinger, “Robust Strategies for Abating Climate Change,” *Climatic Change*, vol. 45, no. 3–4, pp. 387–401, Jun. 2000, doi: 10.1023/A:1005698407365.

[10] S. H. Schneider, "Can We Estimate the Likelihood of Climatic Changes at 2100?," *Climatic Change*, vol. 52, no. 4, pp. 441–451, Mar. 2002, doi: http://dx.doi.org/10.1023/A:1014276210717.

[11] S. Bankes, "Exploratory Modeling for Policy Analysis," *Operations Research*, vol. 41, no. 3, pp. 435–449, Jun. 1993, doi: 10/c7rgcr.

[12] G. G. Garner and K. Keller, "Using Direct Policy Search to Identify Robust Strategies in Adapting to Uncertain Sea-Level Rise and Storm Surge," *Environmental Modelling & Software*, vol. 107, pp. 96–104, Sep. 2018, doi: 10.1016/j.envsoft.2018.05.006.

[13] J. R. Kasprzyk, S. Nataraj, P. M. Reed, and R. J. Lempert, "Many Objective Robust Decision Making for Complex Environmental Systems Undergoing Change," *Environmental Modelling & Software*, vol. 42, pp. 55–71, Apr. 2013, doi: 10.1016/j.envsoft.2012.12.007.